

DSSS Assignment 1

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Introduction and Ethics

The aim of this project is to obtain metadata of articles published across 8 volumes in *Journal of Computational Social Science* and to fit an LDA model with 20 topics based on the abstracts from these articles, in order to analyze the topic of each volume. The data can be obtained either by crawling and scraping all article list pages and the corresponding article URLs to extract metadata from each article's HTML content, or by using the API to directly retrieve metadata for specific article URLs. The LDA modeling and analysis process follows the illustrative example “Abstracts of JSS papers” provided in the `topicmodels` package.

Ethics

Considering ethics, I first checked whether Springer provides an API for accessing article metadata, since using an API is more ethical and legitimate than scraping HTML pages. Fortunately, Springer provides an API for research purposes (I included the API key in my R script). By supplying the article URLs, I was able to access the corresponding article metadata through the API.

Subsequently, I reviewed Springer's Terms of Use. According to the publisher Springer's Terms of Use (Section 1.3), systematically downloading content or continuously and automatically indexing metadata is not permitted. This means it would be inappropriate to use a crawler to collect large amounts of article metadata or to download many PDFs.

I then examined Springer's `robots.txt`. Fortunately, it allows crawlers to access certain article-related resources (e.g., `/article/`, `/article/`, `/articles/`), which means I am allowed to use crawlers to retrieve article URLs.

Therefore, in this study, all metadata used for analysis was accessed via the API by providing the corresponding article URLs.

To ensure compliance with the API's rate limits and terms of use, all requests followed the restrictions of no more than 100 requests per minute and 500 requests per day. The full

metadata was retrieved in JSON format via the API, parsed using regular expressions, and structured into a dataframe for further analysis.

Additionally, for the purpose of demonstrating what I learned in the DSSS course, I used basic web scraping methods (without using the API) to scrape metadata for two articles and download their PDFs. This was done with a clear User-Agent string to identify the educational and research-oriented purpose of the requests, and care was taken to avoid placing any burden on the website.

According to the publisher Springer’s Terms of Use (Section 1.3), systematically downloading content, continuously and automatically searching and indexing meta data is not permitted. However, Springer’s robots.txt allows crawlers to access certain article-related resources (e.g., /article\$, /article/, /articles/\$). Therefore, I only crawled the URLs of the eight article listing pages and scraped two articles to demonstrate the web scraping, element extraction, data binding, and PDF downloading. A 2-second delay was introduced between these requests to avoid overloading the website.

Subsequently, for ethical reasons, the article metadata used in this study was obtained via the API provided by the website for researchers, and all requests complied with the limits of 100 requests per minute and 500 requests per day.

Data and Methods

At the beginning, a headless browser was used to access the homepage of the Journal of Computational Social Science as well as the dynamically loaded article list page and the login page. By controlling the headless browser to perform clicking and input actions, I successfully interacted with multiple webpage elements, achieving login and entry into the article list page. By extracting the pagination information, I determined that the page has a total of 8 pages, containing 377 articles (crawling on May 10, 2025).

Subsequently, the httr package was used to send requests to the URLs of these 8 article list pages. During the request process, a 2-second delay was added to reduce the server load. The response results were saved as RDS files to ensure reproducibility and offline access. The HTML content of each page was parsed using the XML package, and the necessary metadata, such as each article’s URL and DOI, was extracted using XPath.

For ethical considerations, a better way to obtain data is not by scraping web pages but by accessing the API provided by the website. Therefore, before scraping, I checked whether there was a API available to avoid directly scraping the webpage content. Fortunately, I found that Springer provides an API to access metadata. As a result, I used a basic web scraping approach to scrape only two articles to demonstrate the scraping process and included a clear User-Agent in the requests to indicate the purpose of the scraping. Subsequently, I will extract JSON-format data through the API in a structured and legitimate way and use regular expressions to extract the article metadata.

After retrieving the metadata from the API, I consolidated the data into a structured dataframe. This allowed me to store the collected metadata in a CSV file for further analysis. To ensure compliance with ethics and the terms of use, I limited the download to only two articles to demonstrate the download process. Additionally, downloading a large number of PDFs may also result in the IP address being blocked.

Analysis

A mini analysis was conducted on the collected metadata to identify key topics and words across the volumes. First, a topic modeling approach using Latent Dirichlet Allocation (LDA) was applied to the dataset. The analysis generated 20 topics, which allowed for the identification of underlying topics discussed in the articles.

For each volume (from 1 to 8), the most frequent topic was determined, and the top 10 words associated with this topic were extracted. These words represent the most prominent themes for each volume.

Additionally, a TF-IDF (Term Frequency-Inverse Document Frequency) analysis was performed on the article abstracts to highlight distinctive keywords that are significant within individual articles but less common across the entire dataset. This helped to identify unique and important terms.

The results were organized into tables, presenting the top terms for each volume and topic, offering valuable insights into the key concepts across the dataset.

Table 1: Top 10 Words in Most Frequent Topic for 8 Volumes

Volume 1	Volume 2	Volume 3	Volume 4
flow	team	bot	urban
graph	monetari	incom	crime
firm	reward	persuas	dropout
sale	clickbait	debat	climat
incom	judgment	disinform	citi
protest	reput	infodem	crimin
inequ	workflow	parliamentari	blockchain
app	spoiler	journalist	miss
inflat	cgm	affin	satisfact
bdi	flock	african	secur

Table 2: Top 10 Words in Most Frequent Topic for 8 Volumes (Part 2)

Volume 5	Volume 6	Volume 7	Volume 8
diffus	hate	emot	citi
forum	speech	vaccin	llms
stanc	fake	disast	controversi
frame	spr	parti	contagion
toxic	abm	immigr	reddit
bridg	antisemit	antivaccin	gpt
refuge	xai	coupl	tenant
extremist	cancer	fear	cascad
fertil	disast	romant	crowd
stereotyp	slaveri	deepconnect	evacu